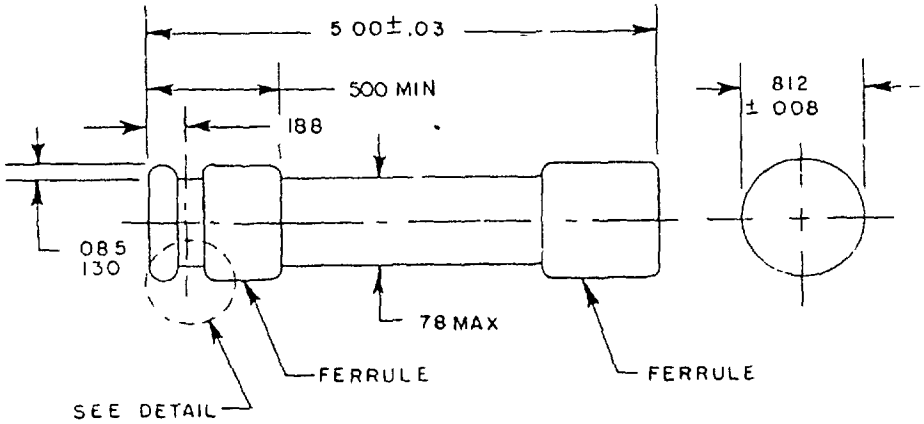
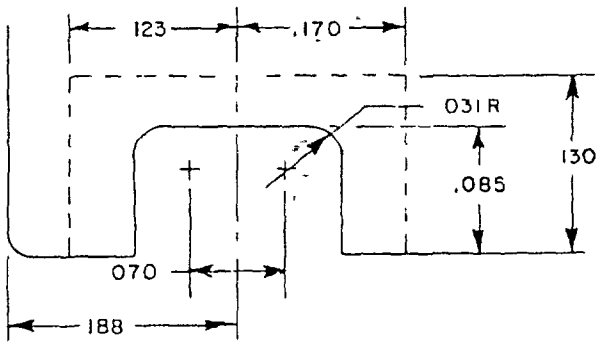
FEDERAL SPECIFICATION SHEET

FUSES, CARTRIDGE, HIGH-INTERRUPTING CAPACITY, CLASSES RK1 AND RK5 (CURRENT-LIMITING) 600 VOLTS, 0-30 AMPERES

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the fuses described herein shall consist of this document and the latest issue in effect of specification W-F-1814/GEN.





REJECTION GROOVE DETAIL

Inches	mm
.008	0.20
.03	0.76
.031	0.79
.070	1.78
.085	2.16
.123	3.12
.130	3.30
.170	4.32
.188	4.79
.500	12.70
.78	19.8
.812	20 62

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3 Solid line indicates location, shape, and dimensions for minimum rejection groove.
4. Dashed line indicates location, shape, and dimensions for maximum rejection groove

FIGURE 1. Classes RK1 and RK5 fuses, 600 volts, 0-30 amperes.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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REQUIREMENTS

Design and construction See figure 1.

Physical Nonrenewable.

Terminals Ferrule type. The ferrules shall be approximately circular in cross-section

Material Brass, copper, or copper alloy.

Strength 5 inch-pound torque between body and ferrules.

Alinement The inside diameter of tubular gage shall be not more than 0.823 inch.

Body

Insulating material Fiber, ceramic, melamine-impregnated glass fiber, or other suitable material.

Electrical

Voltage 600 volts or less.

Frequency 48 to 60 hertz.

Current See table I.

Current carrying capacity 110 percent indefinitely with temperature rise not to exceed 50°C (90°F) above ambient on the body and the ferrules

Overload interrupt

Instantaneous Shall interrupt within 1 hour at 135 percent of rated current and within 2 minutes at 200 percent of rated current.

Time delay In addition to the above requirement, the fuse shall not interrupt 500 percent of rated current within 10 seconds. 1/

Interrupting capacity rating 200,000 amperes rms symmetrical at 600 volts, 48 to 60 hertz and a power factor of 20 percent or less. Closing angle shall be essentially at the zero of the voltage wave (maximum offset) or later, to produce start of arcing within 30 electrical degrees prior to system peak voltage.

Threshold ratio 30 maximum for RK1, 65 maximum for RK5.

Peak let-thru current See table II

Maximum clearing I^2T See table II.

Applicable fuseholder Part number W-F-870/7-03 of W-F-870/7.

Government part number The Government part number shall consist of the prefix "WF1814/62-" followed by the part number designation shown in table I.

1/ A fuse marked "For use only on motor circuits" may clear within 8 minutes at 200 percent.

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TABLE I Current rating and part number designation

Current in amperes	Part number designation			
	RK5		RK1	
	Instantaneous	Time delay	Instantaneous	Time delay
1/10		D5001		D1001
15/100		D50015		D10015
2/10		D5002		D1002
3/10		D5003		D1003
4/10		D5004		D1004
1/2		D5005		D1005
6/10		D5006		D1006
8/10		D5008		D1008
1	F501	D501	F101	D101
1 1/8		D50112		D10112
1 1/4		D50125		D10125
1 4/10		D5014		D1014
1 6/10		D5016		D1016
1 8/10		D5018		D1018
2		D502		D102
2 1/4		D50225		D10225
2 1/2		D5025		D1025
2 8/10		D5028		1028
3	F503	D503	F103	D103
3 2/10		D5032		D1032
3 1/2		D5035		D1035
4		D504		D104
4 1/2		D5045		D1045
5		D505		D105
5 6/10		D5056		D1056
6	F506	D506	F106	D106
6 1/4		D50625		D10625
7		D507		D107
8		D508		D108
9		D509		D109
10	F510	D510	F110	D110
12		D512		D112
15	F515	D515	F115	D115
17 1/2		D5175		D1175
20	F520	D520	F120	D120
25	F525	D525	F125	D125
30	F530	D530	F130	D130

TABLE II. Peak let-thru and I^2T current

	Between threshold and 50 kA		100 kA		200 kA	
	RK1	RK5	RK1	RK5	RK1	RK5
Maximum acceptable peak let-thru current ($I_D \times 10^3$)	6	11	10	11	12	14
Maximum clearing I^2T (amperes squared seconds) ($I^2T \times 10^3$)	10	50	10	50	11	50

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MILITARY INTERESTS

Custodians

Army - ER
Navy - YD
Air Force - 99

Review Activities

Army - AL, AT, MI
Air Force - 99
DLA - ES
NS

User Activities

Army - AR
Navy - OS

CIVIL AGENCY COORDINATING ACTIVITIES

DOE - DOE
COMMERCE - NBS
DOT - ACO, AP1
GPO - GPO
GSA - FBS
JUSTICE - FPI
NASA - JFK, LRC
USPS - POS
VA - OSS

PREPARING ACTIVITY

DLA - ES

(Project 5920-0434-07)

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